Optimization and Curve Sketching

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.02

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Critical points and the first/second derivative tests**

Find the critical points of $f(x) = x^3 - 6x^2 + 9x$, and classify each as a local max or min using the second derivative test.

$f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$, critical points $x = 1, 3$. $f''(x) = 6x - 12$: $f''(1) = -6 < 0$ (local max); $f''(3) = 6 > 0$ (local min).

2 **[Foundational]** • **Target: Critical points and the first/second derivative tests**

For $f(x) = x^4 - 4x^3$, find the intervals of concavity and any inflection points.

$f''(x) = 12x^2 - 24x = 12x(x-2)$. $f'' > 0$ (concave up) on $(-\infty, 0) \cup (2, \infty)$; $f'' < 0$ (concave down) on $(0, 2)$. Inflection points at $x = 0$ and $x = 2$.

3 **[Foundational]** • **Target: Optimization applications**

A farmer has 200 m of fencing to enclose a rectangular field against an existing wall (so only 3 sides need fencing). Find the dimensions that maximize the enclosed area.

Let x = width (two sides), y = length (one side): $2x + y = 200$, so $y = 200 - 2x$. Area $A = xy = x(200 - 2x) = 200x - 2x^2$. $A' = 200 - 4x = 0$ gives $x = 50$, $y = 100$. Maximum area 5000 m^2 at width 50 m, length 100 m.

4 **[Foundational]** • **Target: Optimization applications**

An open-top box is made from a 12 cm square sheet by cutting equal squares of side x from each corner and folding up the sides. Find the value of x that maximizes the box's volume.

$V(x) = x(12 - 2x)^2$, for $0 < x < 6$. $V'(x) = (12-2x)^2 + x \cdot 2(12-2x)(-2) = (12-2x)(12-6x)$. Setting $V' = 0$: $x = 6$ (rejected, gives $V = 0$) or $x = 2$. Maximum volume at $x = 2$ cm.

5 **[Foundational]** • **Target: More optimization**

A cylindrical can must have volume 500π cm^3. Using $V = \pi r^2 h$ and surface area $S = 2\pi r^2 + 2\pi rh$, find the radius that minimizes the material used (surface area).

From V: $h = \frac{500}{r^2}$. $S(r) = 2\pi r^2 + 2\pi r \cdot \frac{500}{r^2} = 2\pi r^2 + \frac{1000\pi}{r}$. $S'(r) = 4\pi r - \frac{1000\pi}{r^2} = 0$: $4r^3 = 1000$, $r^3 = 250$, $r = \sqrt[3]{250} \approx 6.30$ cm.

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6 [Foundational] • Target: More optimization

Find two positive numbers whose sum is 20 and whose product is as large as possible.

Let the numbers be x and $20 - x$: $P(x) = x(20 - x) = 20x - x^2$. $P'(x) = 20 - 2x = 0$ gives $x = 10$. Both numbers are 10, maximum product 100.

7 [Foundational] • Target: Curve sketching

For $f(x) = \frac{1}{3}x^3 - x^2 - 3x + 2$, find the intervals of increase and decrease.

$f'(x) = x^2 - 2x - 3 = (x - 3)(x + 1)$. Critical points $x = -1, 3$. Testing: increasing on $(-\infty, -1)$, decreasing on $(-1, 3)$, increasing on $(3, \infty)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Curve sketching

Using the results above, classify $x = -1$ and $x = 3$ as local maxima or minima via the first derivative test.

f changes from increasing to decreasing at $x = -1$: local maximum. f changes from decreasing to increasing at $x = 3$: local minimum.

9 [Intermediate] • Target: Minimizing distance and cost

Find the point on the parabola $y = x^2$ closest to the point $(0, 3)$, by minimizing $D^2 = x^2 + (x^2 - 3)^2$.

$D^2(x) = x^2 + (x^2 - 3)^2$. $\frac{d}{dx}D^2 = 2x + 2(x^2 - 3)(2x) = 2x[1 + 2(x^2 - 3)] = 2x(2x^2 - 5)$. Setting to 0: $x = 0$ or $x^2 = \frac{5}{2}$. Testing shows $x = 0$ gives a local max (distance 3); $x = \pm\sqrt{5/2}$ gives the minimum, at point $\left(\sqrt{5/2}, \frac{5}{2}\right)$.

10 [Intermediate] • Target: Minimizing distance and cost

A rectangular box with a square base and open top must have volume 32 m^3. Find the base side length that minimizes the surface area, using $S = x^2 + \frac{128}{x}$ (derived from $V = x^2h = 32$).

$S'(x) = 2x - \frac{128}{x^2} = 0$: $2x^3 = 128$, $x^3 = 64$, $x = 4$ m.

11 [Intermediate] • Target: Sketching using derivative information

For $f(x) = 3x^4 - 4x^3$, find all critical points and classify each using the second derivative test.

$f'(x) = 12x^3 - 12x^2 = 12x^2(x - 1)$: critical points $x = 0, 1$. $f''(x) = 36x^2 - 24x$: $f''(0) = 0$ (test inconclusive, needs first derivative test — f' doesn't change sign at $x = 0$, so it's neither a max nor min); $f''(1) = 36 - 24 = 12 > 0$: local minimum at $x = 1$.

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12 [Intermediate] • Target: Sketching using derivative information

For $f(x) = xe^{-x}$, find the critical point and determine whether it is a maximum or minimum.

$f'(x) = e^{-x} - xe^{-x} = e^{-x}(1 - x) = 0$ gives $x = 1$. $f''(x) = -e^{-x}(1 - x) - e^{-x} = e^{-x}(x - 2)$; $f''(1) = e^{-1}(-1) < 0$: local maximum at $x = 1$.

13 [Intermediate] • Target: Visualizing critical points on a graph

The graph shows $f(x) = x^3 - 6x^2 + 9x$ from the first problem. Use the graph to confirm the local max at $x = 1$ and local min at $x = 3$ found algebraically.

The graph rises to a peak at $x = 1$ (local maximum, $f(1) = 4$), then falls to a valley at $x = 3$ (local minimum, $f(3) = 0$), matching the second-derivative-test classification.

14 [Intermediate] • Target: Optimization with a fixed perimeter

A rectangular garden must have a perimeter of 80 m. Find the dimensions that maximize its area.

$2x + 2y = 80$, so $y = 40 - x$. $A(x) = x(40 - x) = 40x - x^2$. $A'(x) = 40 - 2x = 0$ gives $x = 20$, $y = 20$. The maximum-area rectangle is a 20×20 square, area 400 m^2.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Optimization with a fixed perimeter • Cross-Lesson: 16.01 Vector Fields and Line Integrals

A Norman window (a rectangle topped by a semicircle) has perimeter 10 m, with rectangle width x and height y: $2y + x + \pi\frac{x}{2} = 10$. Express the total area $A = xy + \frac{\pi}{8}x^2$ in terms of x alone, and state the equation you would set $A'(x) = 0$ to solve (do not solve).

From the constraint: $y = \frac{10 - x - \frac{\pi x}{2}}{2} = 5 - \frac{x}{2} - \frac{\pi x}{4}$. Substituting: $A(x) = x\left(5 - \frac{x}{2} - \frac{\pi x}{4}\right) + \frac{\pi}{8}x^2 = 5x - \frac{x^2}{2} - \frac{\pi x^2}{4} + \frac{\pi x^2}{8}$. Setting $A'(x) = 5 - x - \frac{\pi x}{2} + \frac{\pi x}{4} = 0$ would give the optimal x.

16 [Challenging] • Target: Concavity and the second derivative test in context • Cross-Lesson: 2.01 Evaluating Limits

A company's profit is modeled by $P(x) = -2x^2 + 80x - 300$, where x is units sold (hundreds). Find the production level that maximizes profit, and verify it is a maximum using the second derivative.

$P'(x) = -4x + 80 = 0$ gives $x = 20$. $P''(x) = -4 < 0$ for all x, confirming a maximum at $x = 20$ (2000 units), with maximum profit $P(20) = -800 + 1600 - 300 = 500$.

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17 [Challenging] • Target: Optimization with a fixed perimeter • Cross-Lesson: 2.02 Continuity

Level 3 · Advanced / AP FRQLet $f(x)$ be a function whose third derivative satisfies $|f'''(x)| \le 6$ for all $x \in [0, 0.4]$. If the second-degree Taylor polynomial $P_2(x)$ centered at $c = 0$ is used to approximate $f(0.2)$, find the maximum possible error in this approximation using the Lagrange Error Bound.

Using the Lagrange Error Bound formula with $n = 2$, $c = 0$, $x = 0.2$, and $M = 6$: $|R_2(0.2)| \le \frac{M}{(2+1)!}|x-c|^{2+1} = \frac{6}{3!}|0.2-0|^3$. Since $3! = 6$: $|R_2(0.2)| \le \frac{6}{6}(0.2)^3 = 1 \times 0.008 = \mathbf{0.008}$ $\left(\text{or } \frac{1}{125}\right)$. The maximum truncation error is guaranteed to be no greater than 0.008.

18 [Challenging] • Target: Concavity and the second derivative test in context • Cross-Lesson: 3.01 The Derivative from First Principles

[Differential Equations · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

19 [Challenging] • Target: Critical points and the first/second derivative tests • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

[Foundations & Precalculus Review · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

20 [Challenging] • Target: Optimization applications • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

[Parametric & Polar · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Critical points and the first/second derivative tests

If $f'(x) = 0$ and $f''(x) = 0$, the Second Derivative Test is:

a. **inconclusive ✓**

b. proves local min

c. proves local max

d. proves inflection

Test fails; need higher derivatives or sign analysis.

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22 [Foundational] • Target: Optimization applications

Second Derivative Test: $f'(c) = 0$ and $f''(c) > 0 \Rightarrow x = c$ is a:

a. local max

b. **local min ✓**

c. inflection point

d. saddle point

$f'' > 0$ at critical point $\Rightarrow$ local min.

23 [Foundational] • Target: More optimization

Use the Second Derivative Test to classify the critical points of $f(x) = 3x^3 - 4x$.

a. $local minimum at x = -\frac{2}{3}, local maximum at x = \frac{2}{3}$

b. $local maximum at x = -\frac{2}{3}, local minimum at x = \frac{2}{3}$ ✓

c. both are local minima

d. both are local maxima

$f'(x) = 3(3)x^2 - 4 = 0 \Rightarrow x = \pm\frac{2}{3}$. Since $f''(x) = 18x$, $f''(-\frac{2}{3})0$ (local minimum).

24 [Foundational] • Target: Curve sketching

If $f'(x) = 0$ and $f''(x) = 0$ at $x = c$, the Second Derivative Test is:

a. guaranteed a maximum

b. guaranteed an inflection point

c. guaranteed a minimum

d. **inconclusive ✓**

When $f''(c) = 0$ too, the test gives no information; another method is needed.

25 [Foundational] • Target: Minimizing distance and cost

Critical points occur where $f'(x)$ is:

a. positive

b. negative

c. **zero or undefined ✓**

d. equal to $f(x)$

Critical points require $f' = 0$ or f' undefined.

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26 [Foundational] • Target: Sketching using derivative information

Second derivative test: $f''(c) > 0$ implies $x = c$ is:

a. local min ✓

b. local max

c. inflection

d. saddle

$f'' > 0$: concave up $\Rightarrow$ local min.

27 [Foundational] • Target: Visualizing critical points on a graph

Consider a function $f(x)$ such that its derivative is given by $f'(x) = 3x^2 - 12x + 10$. The graph of the second derivative, $f''(x)$, is shown below. Given that $f(0) = 5$, find the value of $f(2)$.

a. 15

b. 9 ✓

c. 11

d. 21

The derivative of $f'(x)$ is $f''(x)$. From the graph, we can observe that $f''(x)$ is a parabola opening upwards with its vertex at $(2, -2)$. The equation of a parabola with vertex (h, k) is $a(x-h)^2 + k$. Thus, $f''(x) = a(x-2)^2 - 2$. We can also see that the parabola passes through $(0, 10)$, so $10 = a(0-2)^2 - 2$, which gives $12 = 4a$, so $a = 3$. Therefore, $f''(x) = 3(x-2)^2 - 2 = 3(x^2 - 4x + 4) - 2 = 3x^2 - 12x + 12 - 2 = 3x^2 - 12x + 10$. This matches the given $f'(x)$, confirming our observation of.

28 [Foundational] • Target: Optimization with a fixed perimeter

Which formula gives the length of a polar curve $r = f(\theta)$?

a. $\int_{\alpha}^{\beta} r\, d\theta$

b. $\int_{\alpha}^{\beta} |dr/d\theta|\, d\theta$

c. $\int_{\alpha}^{\beta} r^2\, d\theta$

d. $\int_{\alpha}^{\beta} \sqrt{r^2 + (dr/d\theta)^2}\, d\theta$ ✓

Applying Applications of Integration principles, Polar arc length formula. Therefore, the verified correct option is $\int_{\alpha}^{\beta} \sqrt{r^2 + (dr/d\theta)^2}\, d\theta$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Concavity and the second derivative test in context

For $f(x) = x^4 - 8x^2$, inflection points at $x =$

a. 0 only

b. ± 2

c. $\pm\sqrt{\frac{4}{3}}$ ✓

d. $\pm 2\sqrt{3}$

Setting $f''(x) = 12x^2 - 16 = 0$ gives $x^2 = \frac{4}{3}$, where $f''(x)$ changes sign, so the inflection points occur at $x = \pm\sqrt{\frac{4}{3}}$.

30 [Intermediate] • Target: Critical points and the first/second derivative tests

The second derivative of $f(x) = \sin x$ is:

a. $\cos x$

b. $-\cos x$

c. $-\sin x$ ✓

d. $\sin x$

Applying Derivatives principles, $f' = \cos x$, $f'' = -\sin x$. Therefore, the verified correct option is $-\sin x$.

31 [Intermediate] • Target: Optimization applications

Second derivative of $f = \sin(x)$:

a. $-\sin x$ ✓

b. $\cos x$

c. $-\cos x$

d. $\sin x$

Applying Derivatives principles, $f' = \cos x$, $f'' = -\sin x$. Therefore, the verified correct option is $-\sin x$.

32 [Intermediate] • Target: More optimization

Find the minimum distance from origin to $y = x^2 + 1$:

a. 1

b. $\sqrt{3/4}$

c. $\sqrt{3}/2$ ✓

d. $\sqrt{2}$

Min $D^2 = x^2 + (x^2 + 1)^2$; $dD^2/dx = 2x + 4x(x^2 + 1) = 0 \Rightarrow x = 0$ or $x^2 = -3/4$ (no real). At $x = 0$: $D = 1$.

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33 [Intermediate] • Target: Curve sketching

For polar curve $r = e^{\theta}$, $\theta \in [0, 2\pi]$, the area swept is:

a. $e^{4\pi}$

b. $\frac{1}{2}(e^{4\pi} - 1)$

c. $\frac{1}{4}(e^{4\pi} - 1)$ ✓

d. $\pi e^{4\pi}$

$A = \frac{1}{2}\int_0^{2\pi} e^{2\theta} d\theta = \frac{1}{4}[e^{2\theta}]_0^{2\pi} = \frac{e^{4\pi}-1}{4}$.

34 [Intermediate] • Target: Minimizing distance and cost

To find candidates for the global max/min of f on $[a, b]$, you check critical points and:

a. **the endpoints** a, b ✓

b. inflection points only

c. only $x = 0$

d. points where f is negative

Global extrema on a closed interval occur at critical points or the endpoints.

35 [Intermediate] • Target: Sketching using derivative information

Critical point $x = c$ of f with $f'(c) = 0$: the number of sign changes of f' around c determines:

a. only whether it is max

b. only whether it is min

c. **whether it is max, min, or neither** ✓

d. the concavity

Sign analysis: $+ \to -$ max, $- \to +$ min, no change: neither.

36 [Intermediate] • Target: Visualizing critical points on a graph

Second derivative d^2y/dx^2 parametrically:

a. $\frac{d}{dt}(dy/dx) \cdot \frac{dt}{dx}$ ✓

b. $(d^2y/dt^2)/(d^2x/dt^2)$

c. d^2y/dt^2

d. dy/dx

$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}(y')}{dx/dt}$ where $y' = dy/dx$.

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37 **[Intermediate]** • **Target: Optimization with a fixed perimeter**

Find the inflection points of $f(x) = xe^{-x}$:

a. $x = 1$ only

b. $x = 2$ **only ✓**

c. $x = 0$ only

d. $x = 0$ and $x = 2$

$f'' = e^{-x}(x - 2) = 0 \Rightarrow x = 2$; sign change at $x = 2$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Concavity and the second derivative test in context** • **Cross-Lesson: 3.01 The Derivative from First Principles**

Acceleration is the second derivative of position. If $x(t) = t^3 - 6t^2 + 9t$, when is the particle at rest?

a. $t = 1, 3$ ✓

b. $t = 2$

c. $t = 0, 3$

d. $t = 1, 2$

$v(t) = x'(t) = 3t^2 - 12t + 9 = 3(t - 1)(t - 3) = 0 \Rightarrow t = 1, 3$.

39 **[Challenging]** • **Target: Critical points and the first/second derivative tests** • **Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain**

The method of Lagrange multipliers finds constrained extrema using multiplier λ. The critical value of λ itself:

a. **is the rate of change of the optimal value w.r.t. the constraint constant ✓**

b. gives the extremum value

c. is always zero

d. equals $|ablaf|$

The Lagrange multiplier = sensitivity (shadow price).

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40 [Challenging] • Target: Optimization applications • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Distance particle travels (vs displacement): use:

a. $\int_a^b v(t)dt$

b. $\int_a^b |v(t)|dt$ ✓

c. $v(b) - v(a)$

d. $s(b) - s(a)$

Total distance: integrate speed (absolute velocity).

41 [Challenging] • Target: More optimization • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Find the total distance (not displacement) of $v(t) = \cos(\pi t)$ on $[0, 2]$:

a. 0

b. $2/\pi$

c. 1

d. $4/\pi$ ✓

Speed $= |\cos \pi t|$; $\int_0^2 |\cos \pi t| dt = 4/\pi$.

42 [Challenging] • Target: Curve sketching • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Find all critical points of $f(x) = x^3 - 6x^2 + 9x + 1$.

a. $x = 1$ only

b. $x = 3$ only

c. $x = 1, 3$ ✓

d. $x = 0, 1, 3$

$f' = 3x^2 - 12x + 9 = 3(x - 1)(x - 3) = 0 \Rightarrow x = 1, 3$.

43 [Challenging] • Target: Minimizing distance and cost • Cross-Lesson: 6.01 Related Rates

Analytic function: uniqueness theorem — if f, g analytic on connected open set, and $f = g$ on a sequence of points converging to an interior point, then $f = g$:

a. on the boundary only

b. at all integers

c. at a dense subset

d. **everywhere on the domain** ✓

Identity theorem for analytic functions.

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44 [Challenging] • Target: Sketching using derivative information • Cross-Lesson: 7.01 Integration by Substitution

$f(x) = x^4 - 4x^3$. Concavity: concave up for:

a. $x > 0$

b. $x > 3$ **and** $x < 0$ ✓

c. $x < 0$

d. $x > 2$

$f'' = 12x^2 - 24x = 12x(x-3) > 0$ when $x3$.

45 [Challenging] • Target: Visualizing critical points on a graph • Cross-Lesson: 7.02 Integration by Parts

The length, X mm, of components manufactured by an industrial process is normally distributed with mean μ mm and standard deviation σ mm. A quality assurance engineer tests the null hypothesis $H_0 : \mu = 20.0$ against the alternative hypothesis $H_1 : \mu > 20.0$ at the 5% significance level using an initial random sample of size $n = 25$. The critical region for the sample mean $\bar{X}$ is found to be $\bar{X} \geq 20.658$. Using the standard normal critical value $z_{0.05} = 1.645$: 1. Find the value of σ. 2. Hence, determine the minimum sample size, m, required so that in a subsequent test of $H_0 : \mu = 20.0$ against $H_1 : \mu > 20.0$ at the same 5% significance level, the probability of committing a Type II error when the true mean is $\mu = 21.0$ is at most 0.05.

a. **44** ✓

b. 43

c. 11

d. 22

We solve the problem in two stages: **Step 1: Determine σ** Under H_0, the distribution of the sample mean is $\bar{X} \sim N\left(20.0, \frac{\sigma^2}{25}\right)$. The critical value for a upper one-tailed test at the 5% level corresponds to $z = 1.645$: $\frac{20.658 - 20.0}{\frac{\sigma}{\sqrt{25}}} = 1.645$ $\frac{0.658}{\frac{\sigma}{5}} = 1.645 \Rightarrow \frac{3.29}{\sigma} = 1.645 \Rightarrow \sigma = \frac{3.29}{1.645} = 2.0$ mm **Step 2: Determine the minimum sample size m** For.

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